Geometric-Scattering Stability to Diffeomorphisms Was Proved in the Expanded Paper

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17 August 2026

Status: literature already answered (qualified affirmative answer; no new theorem is claimed here).

The 2018 target

Perlmutter, Wolf, and Hirn introduce geometric scattering on compact smooth Riemannian manifolds. After proving commutator estimates for its spectral filters, their introduction says these results should permit a future study of stability of the *scattering network* to diffeomorphisms, using a notion analogous to Euclidean Lipschitz stability [1] (source PDF page 3).

This is more than generic future-work language: it identifies the missing upgrade from individual-filter control to stability of the nonlinear cascade.

Explicit later answer

The authors' expanded paper with Gao, arXiv:1905.10448, carries out that upgrade [2]. Let S_J be the windowed geometric wavelet scattering transform on a compact d-manifold $\mathcal{M}$, let $V_\zeta f = f \circ \zeta^{-1}$, and suppose f is λ-bandlimited. Theorem 4.1 (supporting PDF pages 12–13) proves that, when a diffeomorphism is factored into an isometric part ζ_1 and a residual part ζ_2,

$$\|S_J f - S_J V_\zeta f\|_{2,2} \leq C(\mathcal{M}) \Big[2^{-dJ} \|\zeta_1\|_\infty \|U_J f\|_2 + \lambda^d \|\zeta_2\|_\infty \|f\|_2 \Big] .$$

It also gives the finite-depth version, with the first term multiplied by $(L+1)^{1/2}$ and $\|f\|_2$ factoring the entire bound.

Theorem 4.2 (supporting PDF page 13) sends the averaging scale to infinity and obtains the non-windowed estimate

$$\|\overline{S} f - \overline{S} V_\zeta f\|_2 \leq \lambda^d \|\zeta_2\|_\infty \|f\|_2.$$

Thus the specific network-level diffeomorphism-stability program from 2018 received a quantitative affirmative answer.

Idea of the later proof

Bandlimiting makes deformation of the input controllable:

$$\|f - V_\zeta f\|_2 \leq C(\mathcal{M}) \lambda^d \|\zeta\|_\infty \|f\|_2.$$

The later authors split the deformation into an isometry and a residual diffeomorphism. Local isometry invariance controls the first part, while nonexpansiveness of the scattering cascade propagates the input estimate for the second. Combining the two estimates yields Theorem 4.1; taking $J \to \infty$ removes the window and yields Theorem 4.2.

Scope of the answer

The answer is qualified. Theorems 4.1–4.2 assume bandlimited input. The later Section 4.2 obtains an unbandlimited result for *finite-width* networks on two-point homogeneous manifolds, but explicitly leaves the general infinite-width problem open. It also leaves future comparison of scattering coefficients on two manifolds that are diffeomorphic but not isometric. These limitations do not undo the affirmative answer to the 2018 program; they delimit it precisely.

References

[1] M. Perlmutter, G. Wolf, and M. Hirn, *Geometric Scattering on Manifolds*, arXiv:1812.06968 (2018).

[2] M. Perlmutter, F. Gao, G. Wolf, and M. Hirn, *Geometric Wavelet Scattering Networks on Compact Riemannian Manifolds*, arXiv:1905.10448; Proceedings of Machine Learning Research **145** (2021), 570–604.